Reference-paper dose-response dataset

Simultaneous statistical inferences of treatment–control mean differences

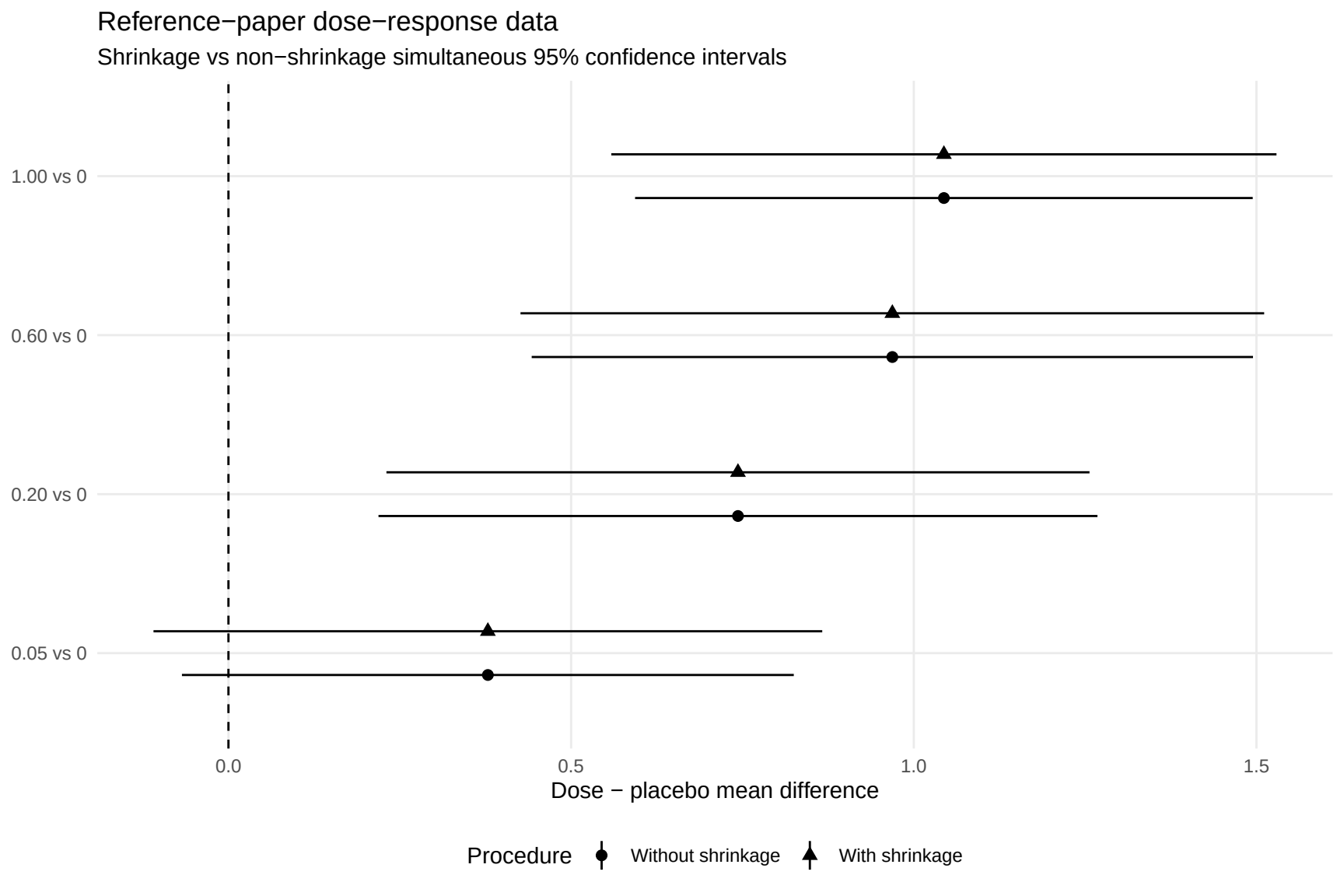
Treatment–control comparison	Point estimate (treatment − control)	95% simultaneous confidence interval	Interval length
Proposed procedure without shrinkage			
0.05 vs 0	0.379	(-0.068, 0.825)	0.893
0.20 vs 0	0.743	(0.219, 1.268)	1.049
0.60 vs 0	0.969	(0.442, 1.495)	1.052
1.00 vs 0	1.044	(0.593, 1.495)	0.901
Proposed procedure with shrinkage			
0.05 vs 0	0.379	(-0.109, 0.866)	0.976
0.20 vs 0	0.743	(0.231, 1.256)	1.026
0.60 vs 0	0.969	(0.426, 1.511)	1.085
1.00 vs 0	1.044	(0.559, 1.529)	0.971

Original 100-patient dataset; five arms of 20 patients; 32 original missing responses retained. All responses and missing-value locations were verified against published Table 5.

One application analysis with 40 treatment-restricted approximate Bayesian bootstrap (ABB) imputations. Shrinkage and non-shrinkage share the same imputations and point estimates. These are actual simultaneous 95% intervals, not averages over repeated simulations.

This script uses within-dose ABB rather than regression because the published dataset contains no auxiliary covariate. The original reference paper reported 20 propensity-score imputations. Its reported Table 6 values are stored separately and are not results computed by this script.

Conclusion: dose 0.05 is not significant; doses 0.20, 0.60 and 1.00 are significant under both procedures. Shrinkage shortens the interval only for dose 0.20; the other intervals become longer.


Reference−paper dose−response data
Shrinkage vs non−shrinkage simultaneous 95% confidence intervals
1.00 vs 0
0.60 vs 0
0.20 vs 0
0.05 vs 0
0.0
0.5
1.0
1.5
Dose − placebo mean difference
Procedure
Without shrinkage
With shrinkage